Robot Control II

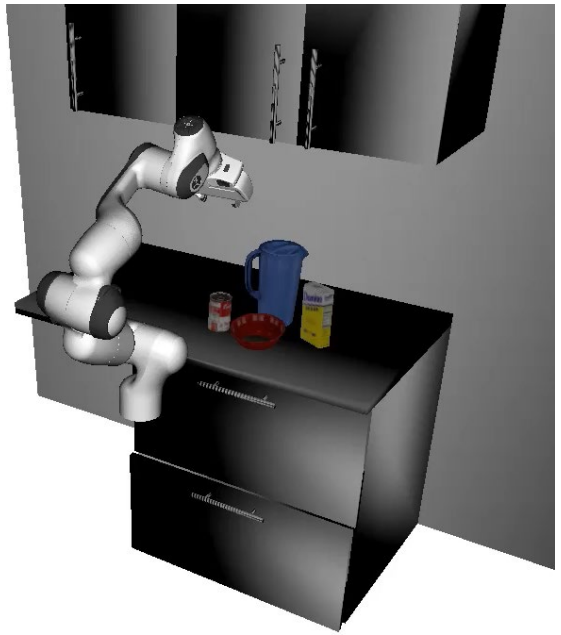
CS 6341 Robotics

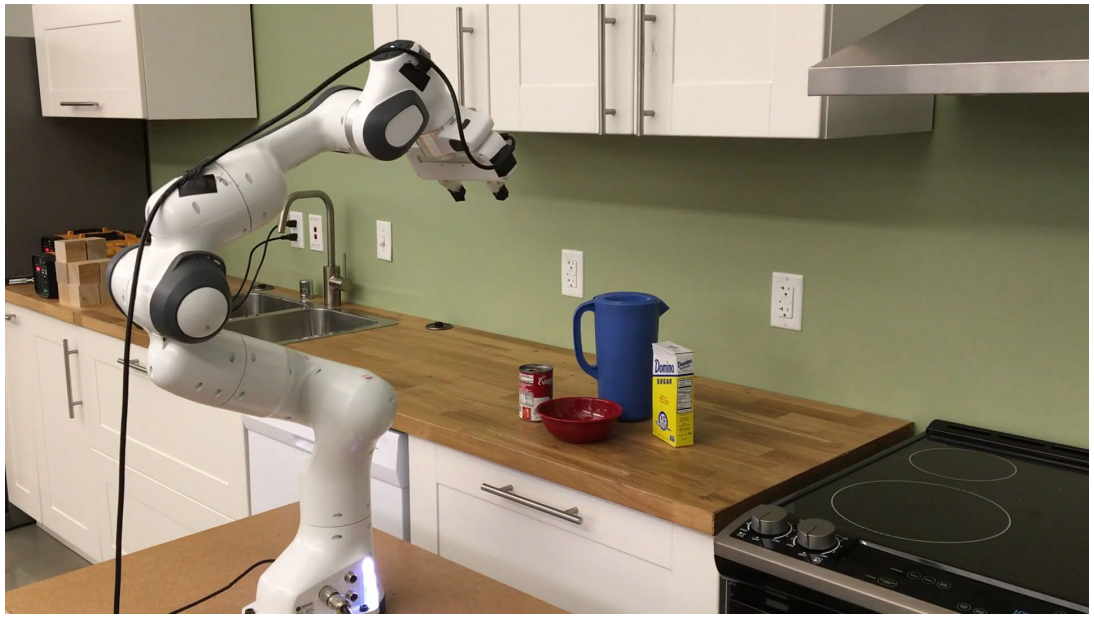
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Motion Control

- Goal: follow a given robot trajectory





Motion Control

- Typically, we assume direct control of the forces or torques at robot joints

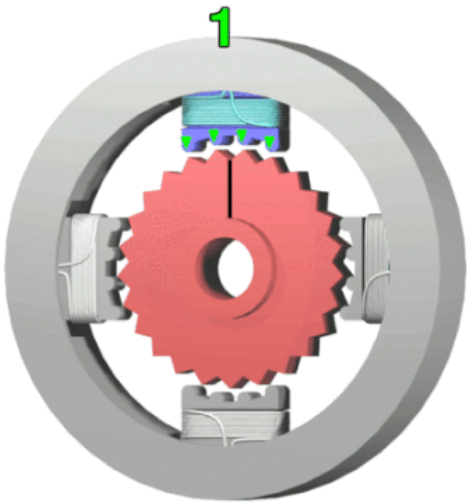


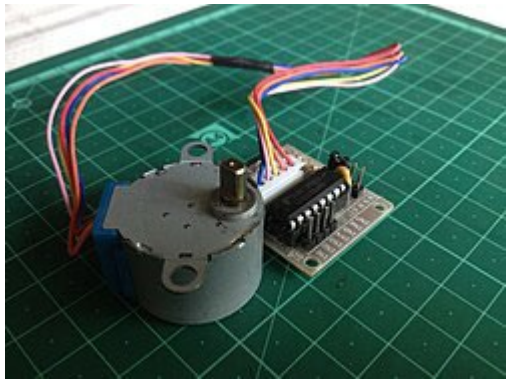
- In some cases, we can assume that there is direct control of the joint velocities



Stepper Motors

- The velocity of a joint is determined directly by the frequency of the pulse train sent to the stepper motor https://en.wikipedia.org/wiki/Stepper_motor





- A **stepper motor** moves in **discrete angular steps** (e.g., 1.8° per step)
- Send **step pulses** at a certain rate (say, 1000 pulses per second)
- Angular velocity

$$\omega = (\text{step angle}) \times (\text{pulse rate})$$

$$\omega = 1.8^\circ \times 1000 = 1800^\circ/s = 5 \text{ revolutions per second}$$

Stepper motors are best for low-speed, precise motion, but not ideal for high-speed or high-torque robotic applications

Motion Control of a Single Joint

- Proportional controller or P controller

Control rule

$$\dot{\theta}(t) = K_p(\theta_d(t) - \theta(t)) = K_p\theta_e(t)$$

Control gain

$$K_p > 0$$

- When $\theta_d(t)$ is a constant $\dot{\theta}_d(t) = 0$

Setpoint control

  - Error dynamics

$$\dot{\theta}_e(t) = \cancelto{0}{\dot{\theta}_d(t)} - \dot{\theta}(t)$$

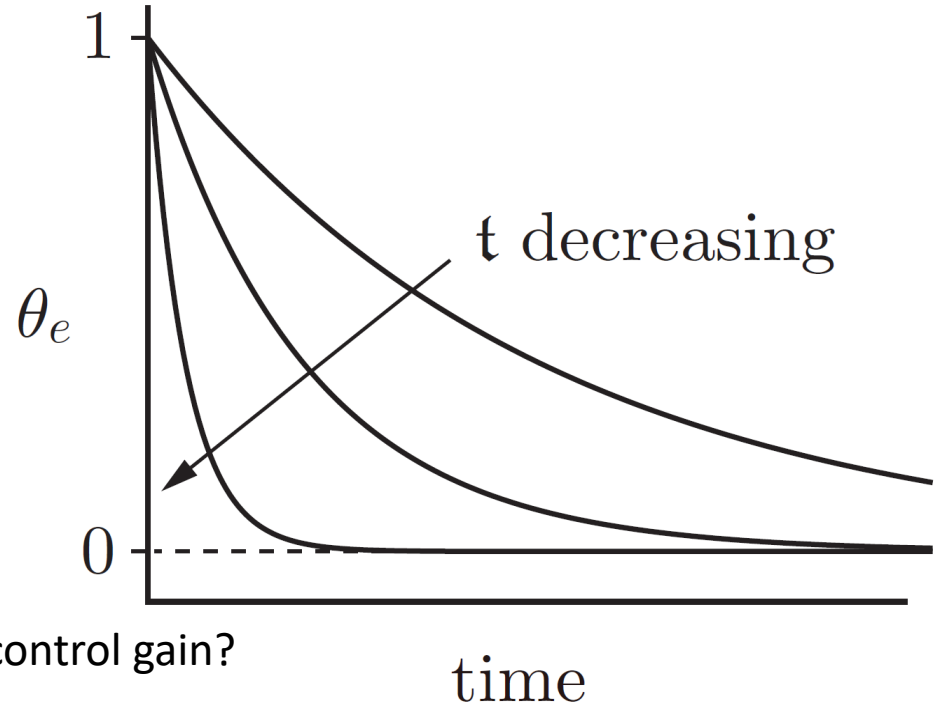
$$\dot{\theta}_e(t) = -K_p\theta_e(t) \quad \rightarrow \quad \dot{\theta}_e(t) + K_p\theta_e(t) = 0$$

First-Order Error Dynamics

First-Order Error Dynamics

$$\dot{\theta}_e(t) + \frac{1}{\mathfrak{t}}\theta_e(t) = 0 \qquad \text{time constant } \mathfrak{t}$$

Solution $\theta_e(t) = e^{-t/\mathfrak{t}}\theta_e(0)$



Setpoint control

$$\dot{\theta}_e(t) + K_p\theta_e(t) = 0 \qquad \mathfrak{t} = 1/K_p$$

- 0 steady state error
- No overshoot
- 2% settling time $4/K_p$

How shall we choose the control gain?

Larger K_p is better

P Controller

$$\dot{\theta}(t) = K_p(\theta_d(t) - \theta(t)) = K_p\theta_e(t)$$

- When $\theta_d(t)$ is not constant but $\dot{\theta}_d(t)$ is constant $\dot{\theta}_d(t) = c$
- Error dynamics

velocity setpoint control

$$\dot{\theta}_e(t) = \dot{\theta}_d(t) - \dot{\theta}(t) = c - K_p\theta_e(t)$$

Solution

$$\theta_e(t) = \frac{c}{K_p} + \left(\theta_e(0) - \frac{c}{K_p}\right)e^{-K_p t} \longrightarrow c/K_p$$

steady-state error

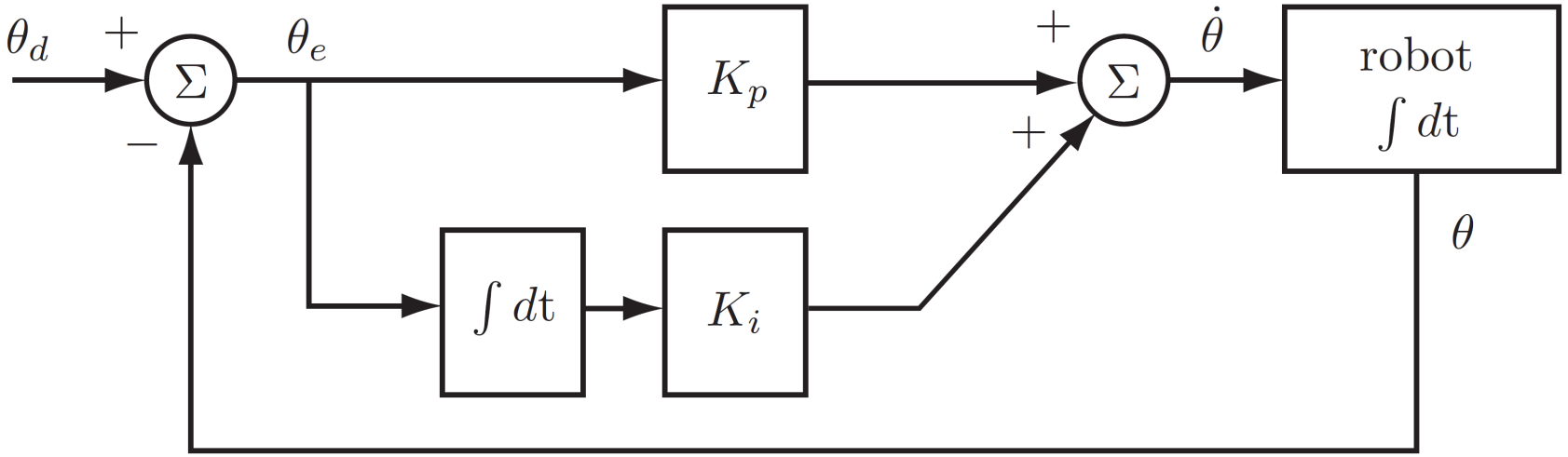
We cannot make K_p arbitrarily large (velocity limit, instability)

PI Controller

- A proportional-integral controller

Control rule $$\dot{\theta}(t) = K_p\theta_e(t) + K_i \int_0^t \theta_e(\mathrm{t})\, d\mathrm{t}$$

Time-integral of the error



PI Controller

- Error dynamics for a constant $\dot{\theta}_d(t) = c$ velocity setpoint control

$$\dot{\theta}(t) = K_p\theta_e(t) + K_i \int_0^t \theta_e(\mathrm{t})\, d\mathrm{t}$$

$$\dot{\theta}_e(t) = \dot{\theta}_d(t) - \dot{\theta}(t)$$

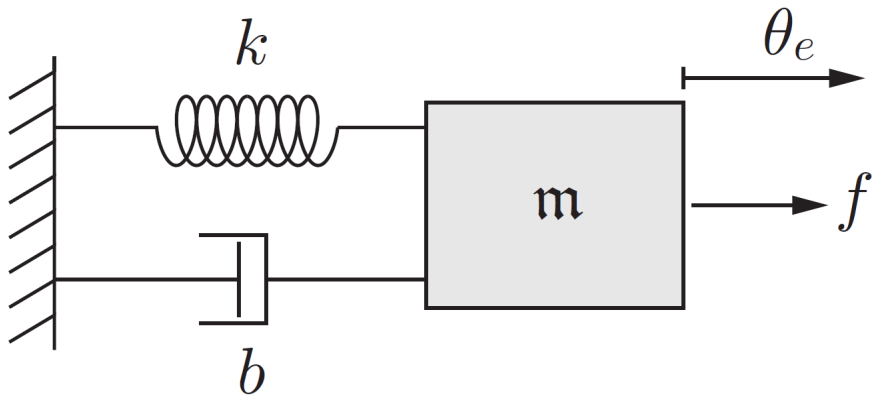
$$\dot{\theta}_e(t) + K_p\theta_e(t) + K_i \int_0^t \theta_e(\mathrm{t})\, d\mathrm{t} = c$$

$$\ddot{\theta}_e(t) + K_p\dot{\theta}_e(t) + K_i\theta_e(t) = 0$$ Second-Order Error Dynamics

PI Controller

- Mass-spring-damper

$$\mathfrak{m}\ddot{\theta}_e + b\dot{\theta}_e + k\theta_e = f$$



Natural frequency $\omega_n = \sqrt{\frac{k}{m}}$

Damping ratio $\zeta = \frac{b}{2\sqrt{km}}$ (zeta)

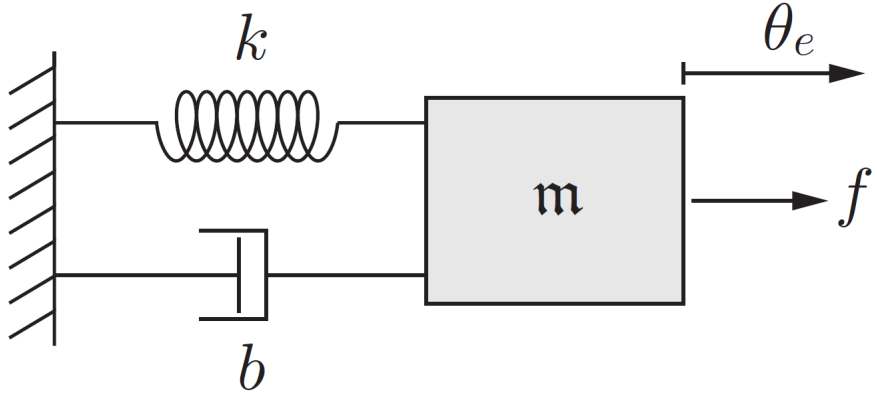
$$\frac{k}{m} = \omega_n^2$$

$$\frac{b}{m} = 2\zeta\omega_n$$

$$f = 0 \qquad \ddot{\theta}_e(t) + \frac{b}{\mathfrak{m}}\dot{\theta}_e(t) + \frac{k}{\mathfrak{m}}\theta_e(t) = 0$$

PI Controller

$$\ddot{\theta}_e(t) + \frac{b}{\mathfrak{m}}\dot{\theta}_e(t) + \frac{k}{\mathfrak{m}}\theta_e(t) = 0$$



$$\ddot{\theta}_e(t) + 2\zeta\omega_n\dot{\theta}_e(t) + \omega_n^2\theta_e(t) = 0$$

Standard second-order form

natural frequency ω_n

damping ratio ζ

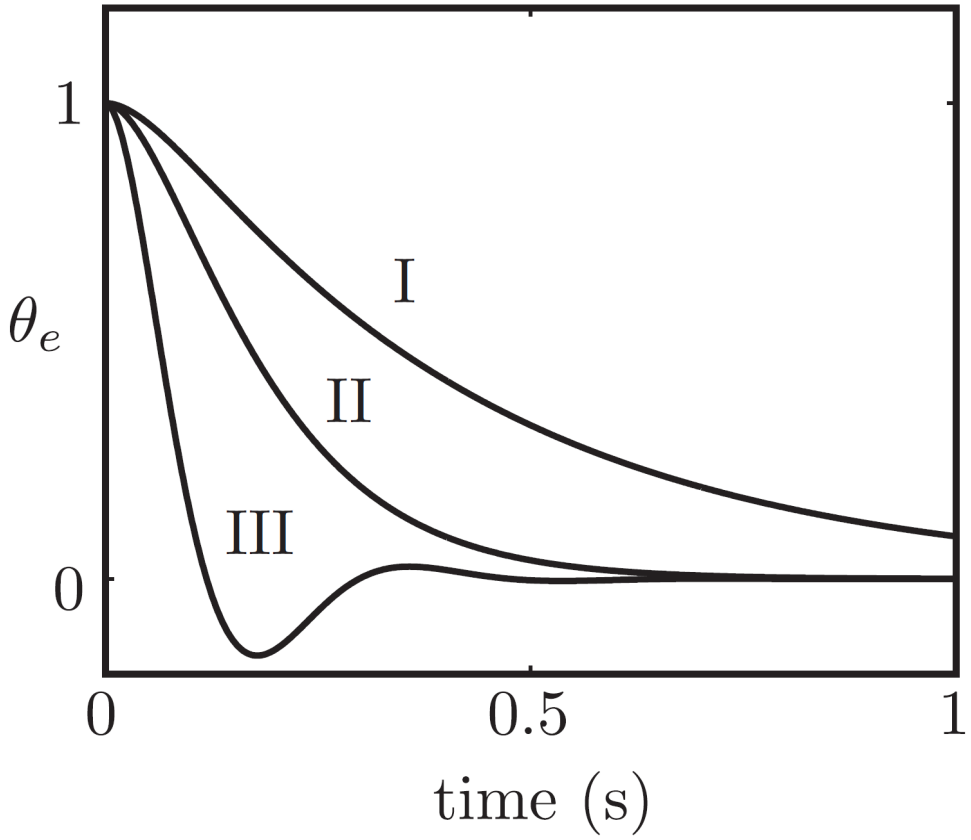
$$\omega_n = \sqrt{K_i}$$

$$\ddot{\theta}_e(t) + K_p\dot{\theta}_e(t) + K_i\theta_e(t) = 0$$

$$\zeta = K_p/(2\sqrt{K_i})$$

PI Controller

Error response



Damping Ratio (ζ)	Behavior	Motion	Damped Frequency
$\zeta = 0$	Undamped	Pure oscillation	ω_n
$0 < \zeta < 1$	Underdamped	Oscillates and decays	$\omega_n\sqrt{(1-\zeta^2)}$
$\zeta = 1$	Critically damped	No oscillation, fastest return	0
$\zeta > 1$	Overdamped	No oscillation, slow return	*Imaginary* (no real ω_n)

$$K_p = 20 \qquad \zeta = K_p/(2\sqrt{K_i})$$

- Overdamped $\zeta = 1.5,\ K_i = 44.4,$ case I
- Critically damped $\zeta = 1,\ K_i = 100,$ case II
- Underdamped $\zeta = 0.5,\ K_i = 400,$ case III

Which one is the best?

Motion Control with Velocity Inputs

- Motion control with velocity inputs
  - Given a desired trajectory of a robot in joint space or in task space

$$\theta_d(t) \qquad X_d(t)$$

- Proportional controller or P controller

$$\dot{\theta}(t) = K_p(\theta_d(t) - \theta(t)) = K_p\theta_e(t)$$

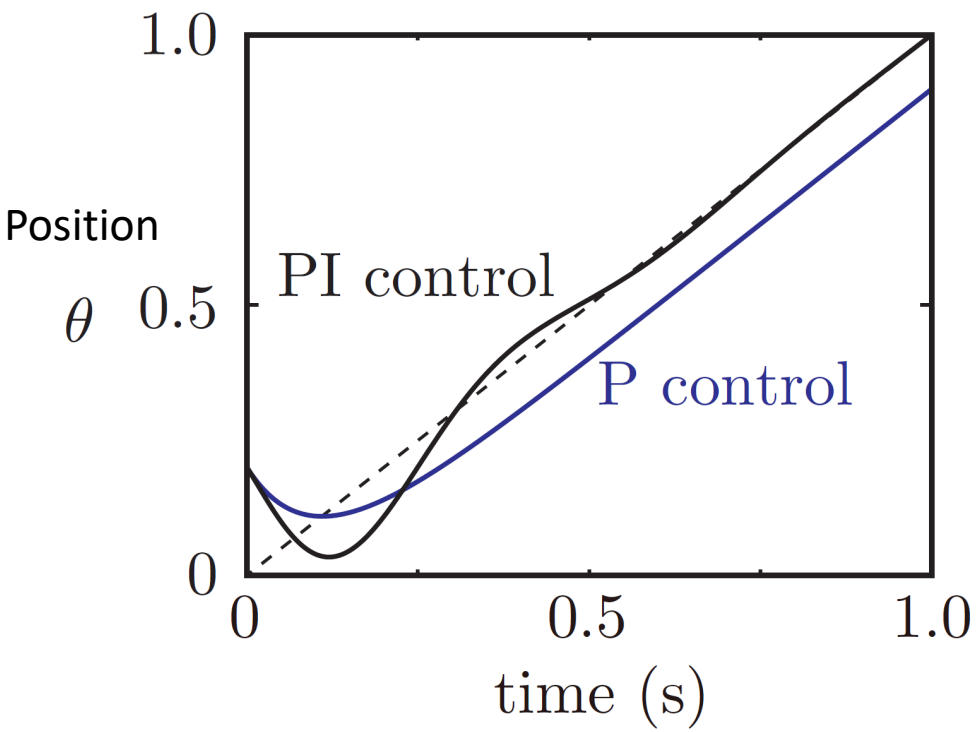
Control gain $K_p > 0$

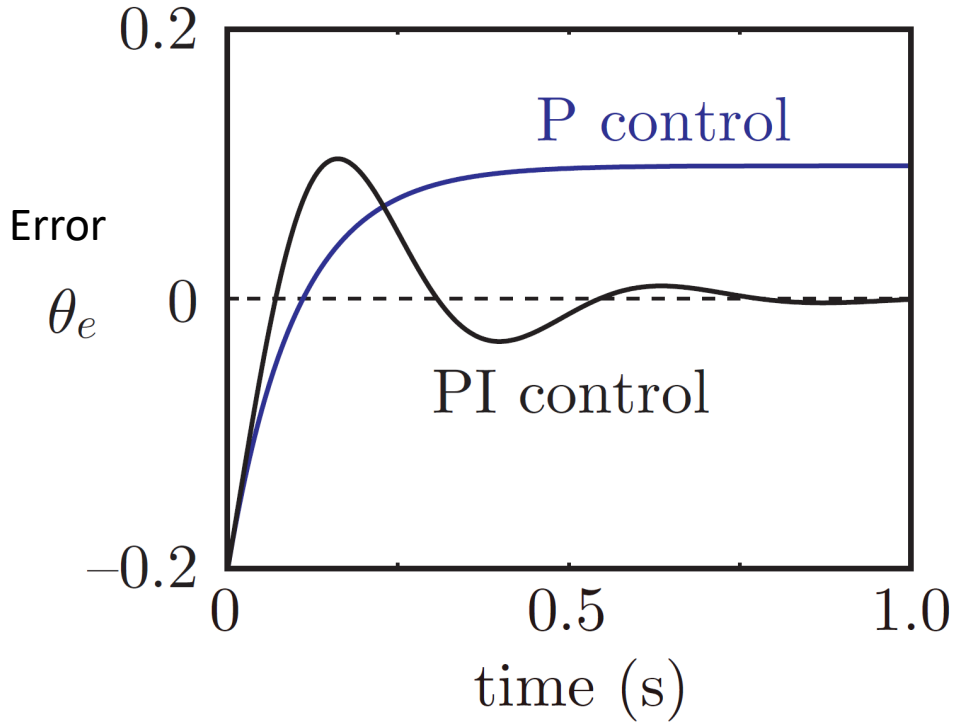
- Proportional-integral controller or PI controller

$$\dot{\theta}(t) = K_p\theta_e(t) + K_i \int_0^t \theta_e(\mathrm{t})\, d\mathrm{t}$$

Comparison between P Controller and PI Controller

$$\dot{\theta}_d(t) = c$$ velocity setpoint control





Reference trajectory (dashed)

Feedforward Plus Feedback Control

- Feedback control: an error is required before the joint begins to move

$$\dot{\theta}(t) = K_p\theta_e(t) + K_i \int_0^t \theta_e(\mathrm{t})\, d\mathrm{t}$$

- Feedforward plus feedback control: Initiate motion before any error accumulates

$$\dot{\theta}(t) = \dot{\theta}_d(t) + K_p\theta_e(t) + K_i \int_0^t \theta_e(\mathrm{t})\, d\mathrm{t}$$

$$\dot{\theta}_e(t) = \dot{\theta}_d(t) - \dot{\theta}(t)$$

$$\ddot{\theta}_e(t) + K_p\dot{\theta}_e(t) + K_i\theta_e(t) = 0$$

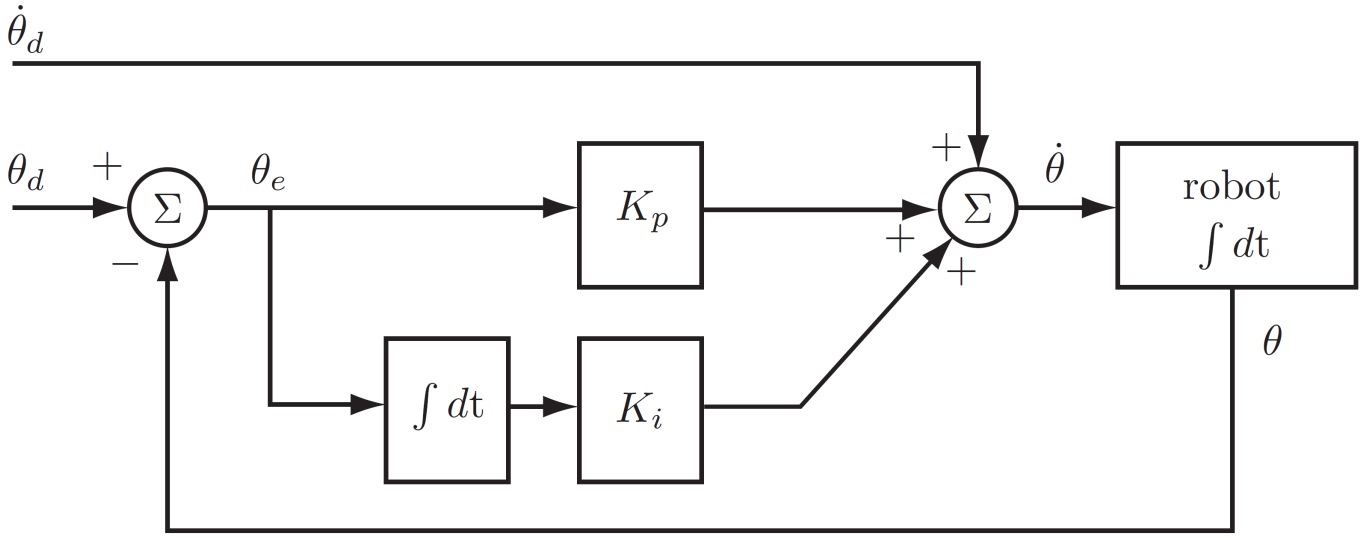
The same error dynamics as the feedback PI controller

Incorporate the reference joint velocity

Feedforward Plus Feedback Control

- Feedforward plus feedback control: Initiate motion before any error accumulates

$$\dot{\theta}(t) = \dot{\theta}_d(t) + K_p\theta_e(t) + K_i \int_0^t \theta_e(\mathrm{t})\, d\mathrm{t}$$



Preferred control law for producing a commanded velocity to the joint

Motion Control of Multi-Joint Robots

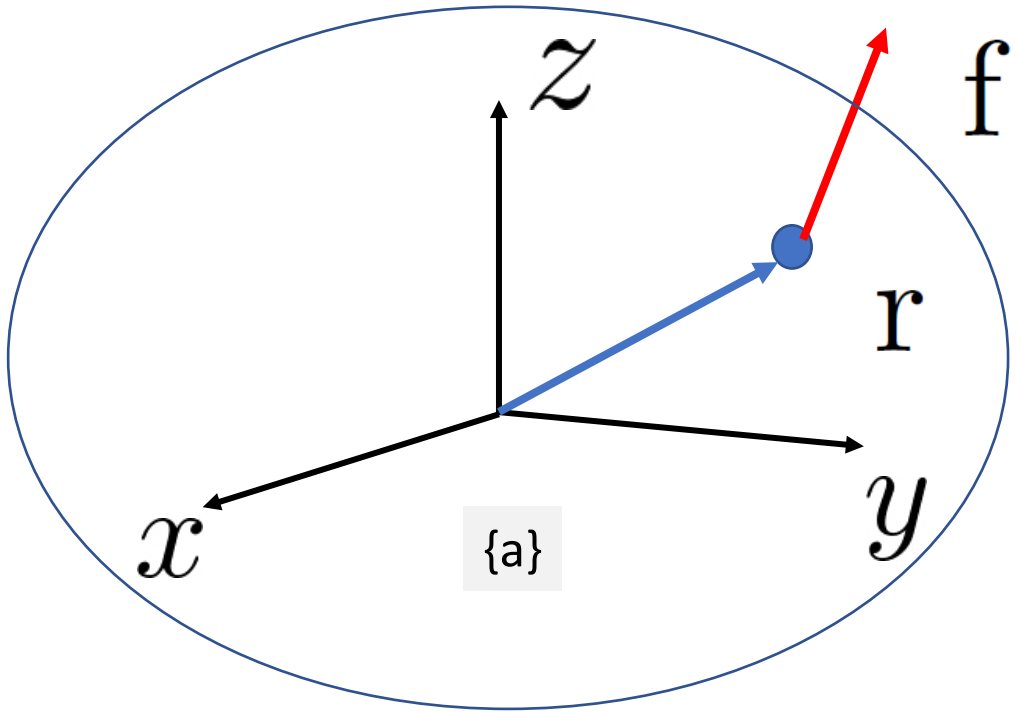
- Reference position $\theta_d(t)$ and actual position $\theta(t)$ n dimensional vector

- Gains $K_p\ K_i\ \ n \times n$ matrix

$$k_p I \qquad k_i I$$

Control law $$\dot{\theta}(t) = \dot{\theta}_d(t) + K_p\theta_e(t) + K_i \int_0^t \theta_e(\mathrm{t})\ d\mathrm{t}$$

Torque



A rigid body

Point $r_a \in \mathbb{R}^3$

Force $f_a \in \mathbb{R}^3$

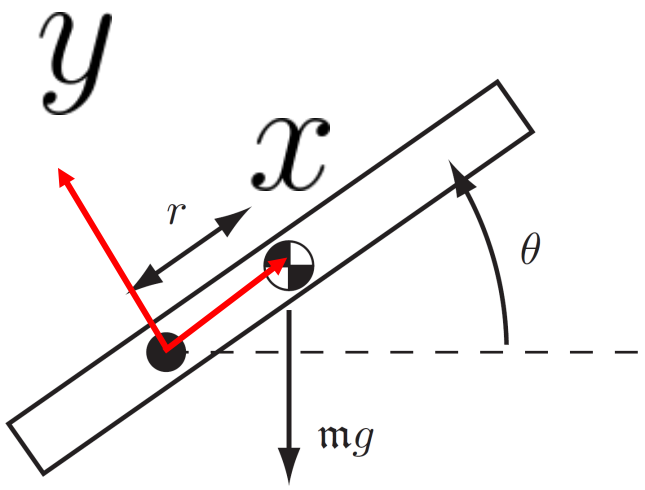
Torque or Moment

$$m_a \in \mathbb{R}^3$$

$$m_a = r_a \times f_a$$

Motion Control with Torque or Force Inputs

- Motion Control of a single joint



Dynamics $\tau = M\ddot{\theta} + \mathfrak{m}gr\cos\theta + b\dot{\theta}$

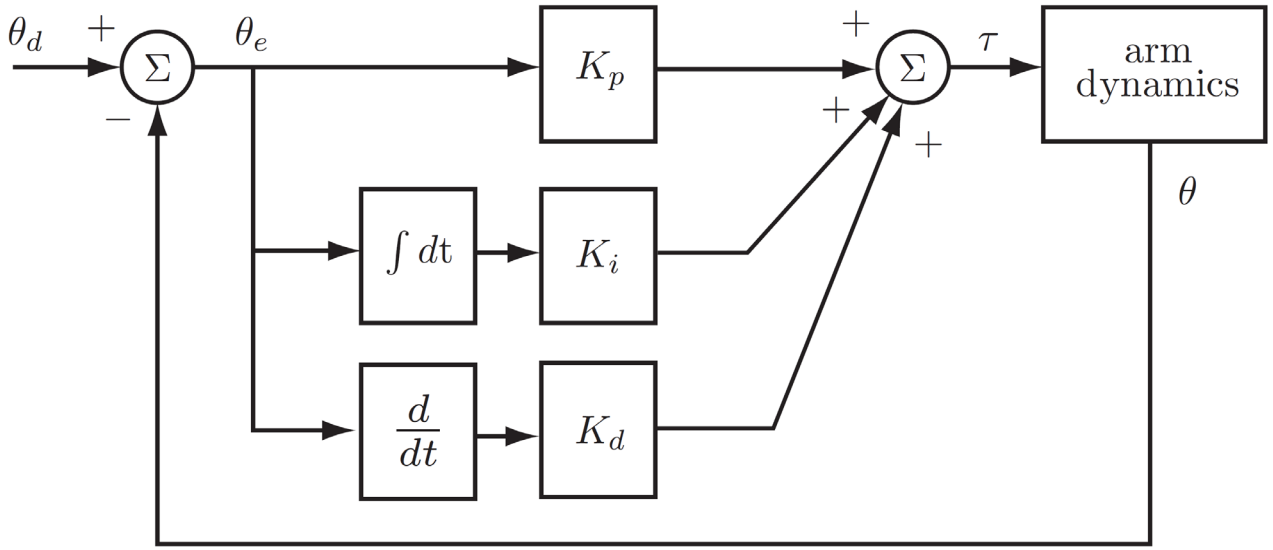
Scalar inertia mass

$$\tau = M\ddot{\theta} + h(\theta, \dot{\theta})$$

Motion Control of a Single Joint

- Feedback control: PID control
  - Proportional-Integral-Derivative control

$$\tau = K_p\theta_e + K_i \int \theta_e(\mathrm{t})d\mathrm{t} + K_d\dot{\theta}_e \qquad \theta_e = \theta_d - \theta$$



PD Control

- Dynamics $\tau = M\ddot{\theta} + \mathfrak{m}gr\cos\theta + b\dot{\theta}$
- PD control law $K_p(\theta_d - \theta) + K_d(\dot{\theta}_d - \dot{\theta})$

Assume $g = 0$

$$M\ddot{\theta} + b\dot{\theta} = K_p(\theta_d - \theta) + K_d(\dot{\theta}_d - \dot{\theta})$$

Control the torque

Control objective: constant θ_d $\dot{\theta}_d = \ddot{\theta}_d = 0$

Setpoint control

$$\theta_e = \theta_d - \theta \quad \dot{\theta}_e = -\dot{\theta} \quad \ddot{\theta}_e = -\ddot{\theta}$$

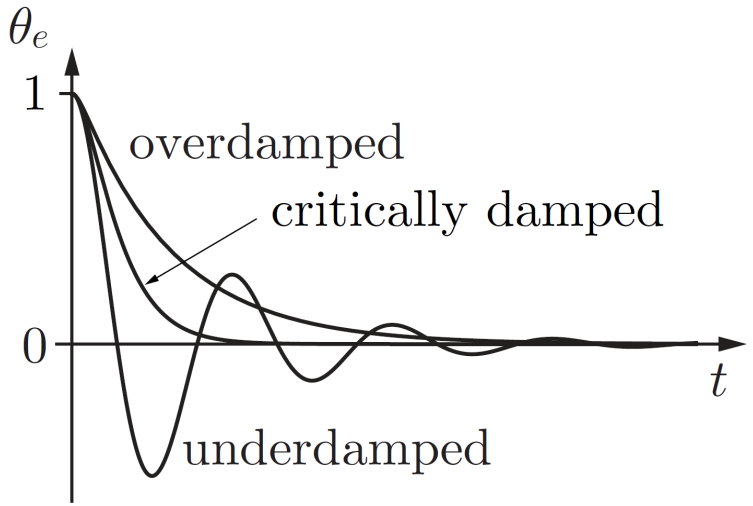
Error dynamics $M\ddot{\theta}_e + (b + K_d)\dot{\theta}_e + K_p\theta_e = 0$

PD Control

- Standard second-order form

$$M\ddot{\theta}_e + (b + K_d)\dot{\theta}_e + K_p\theta_e = 0$$

$$\ddot{\theta}_e + \frac{b + K_d}{M}\dot{\theta}_e + \frac{K_p}{M}\theta_e = 0 \quad \rightarrow \quad \ddot{\theta}_e + 2\zeta\omega_n\dot{\theta}_e + \omega_n^2\theta_e = 0$$



$$\zeta = \frac{b + K_d}{2\sqrt{K_p M}} \qquad \omega_n = \sqrt{\frac{K_p}{M}}$$

Critically damped: $\zeta = 1$

PD Control

- When g > 0, the error dynamics

$$M\ddot{\theta}_e + (b + K_d)\dot{\theta}_e + K_p\theta_e = \mathfrak{m}gr\cos\theta$$

When the joint comes to rest at a configuration θ ,

$$\dot{\theta} = \ddot{\theta} = 0$$

$$K_p\theta_e = \mathfrak{m}gr\cos\theta$$

$$\text{the final error } \theta_e \text{ is nonzero when } \theta_d \neq \pm\pi/2$$

Non-zero steady-state error

PID Control

- Setpoint error dynamics

$$M\ddot{\theta}_e + (b + K_d)\dot{\theta}_e + K_p\theta_e + K_i \int \theta_e(\mathrm{t})d\mathrm{t} = \tau_{\mathrm{dist}}$$

Disturbance torque $\mathrm{m}gr\cos\theta$

Taking derivatives

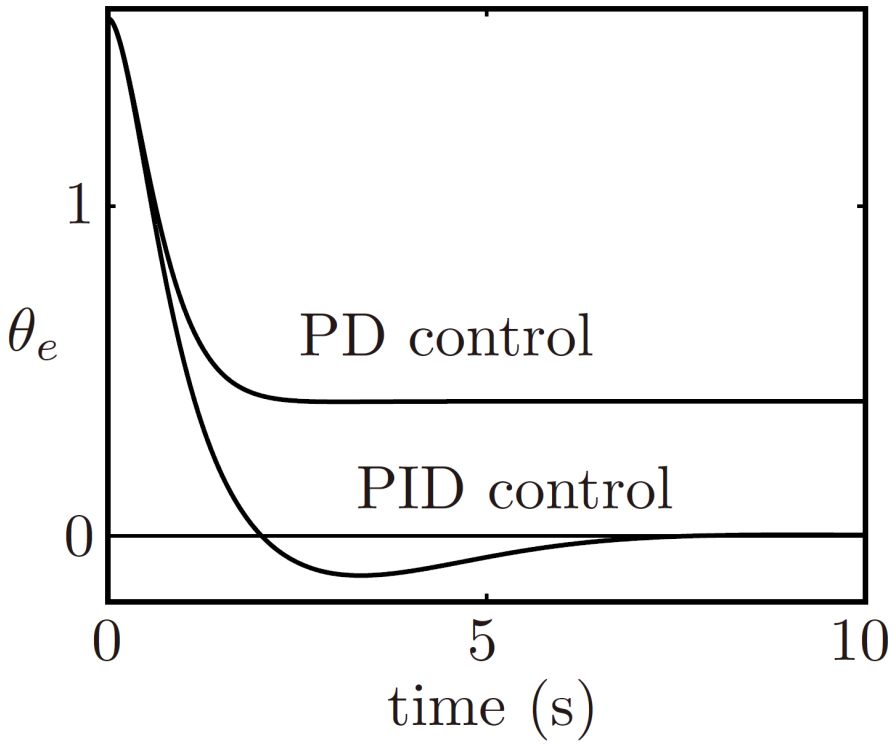
$$M\theta_e^{(3)} + (b + K_d)\ddot{\theta}_e + K_p\dot{\theta}_e + K_i\theta_e = \dot{\tau}_{\mathrm{dist}}$$

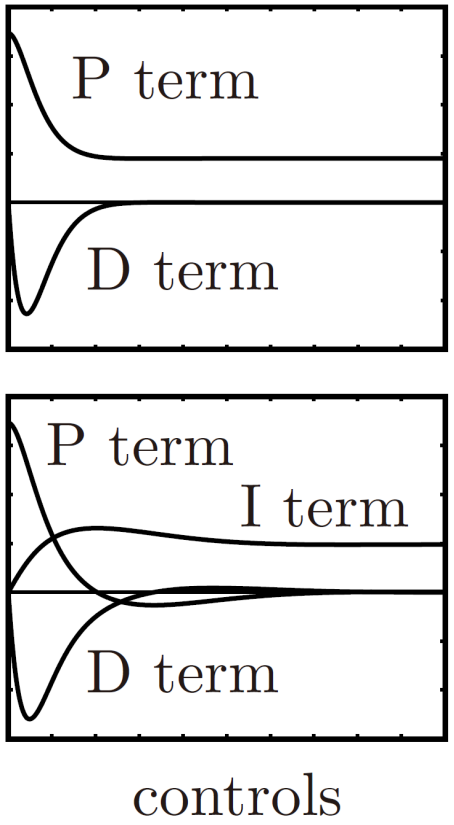
Third-Order Error Dynamics $$s^3 + \frac{b + K_d}{M}s^2 + \frac{K_p}{M}s + \frac{K_i}{M} = 0$$

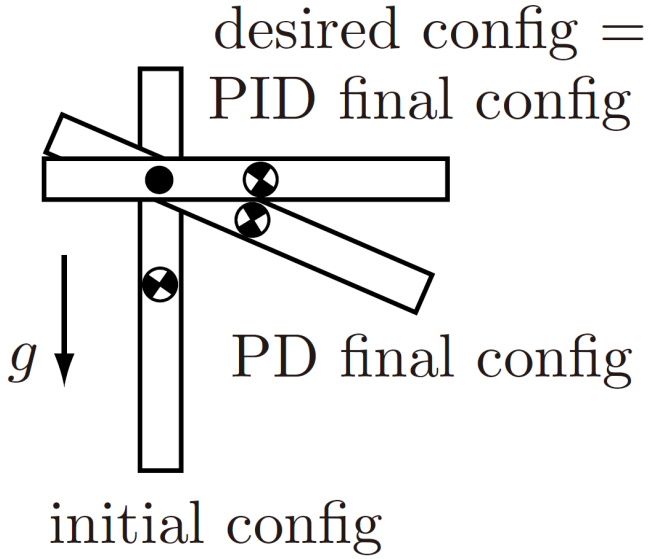
If τ_{dist} Constant

If all roots have a negative real part, then the error dynamics is stable, and θ_e converges to zero

PID Control







$$\theta(0) = -\pi/2, \dot{\theta}(0) = 0$$

$$\theta_d = 0, \dot{\theta}_d = 0$$

PID Control

```
time = 0                              // dt = servo cycle time
eint = 0                              // error integral
qprev = senseAngle                    // initial joint angle q
loop
  [qd,qdotd] = trajectory(time) // from trajectory generator

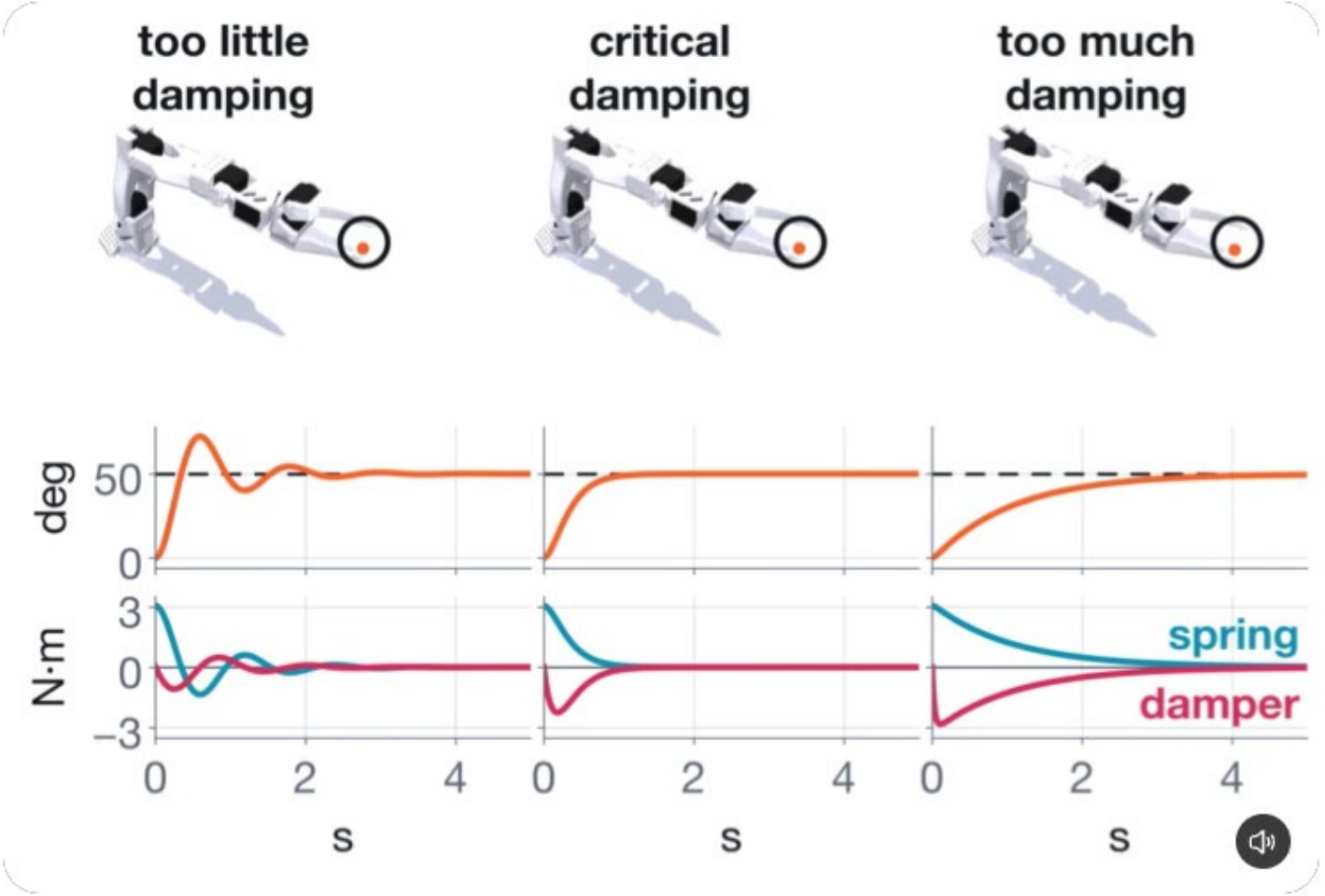
  q = senseAngle                      // sense actual joint angle
  qdot = (q - qprev)/dt               // simple velocity calculation
  qprev = q

  e = qd - q
  edot = qdotd - qdot
  eint = eint + e*dt

  tau = Kp*e + Kd*edot + Ki*eint
  commandTorque(tau)

  time = time + dt
end loop
```

SO101 Control Example



https://x.com/ja_rothschild/status/2100633491432239411

Summary

- Motion control with velocities
  - P controller
  - PI controller
- Motion control with torque or force Inputs
  - PD controller
  - PID controller

Further Reading

- Chapter 11 in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017.